

A Robust Bread Defect Detection and Counting System

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Abstract — In this paper, an automated visual inspection method is presented to detect defect and count breads for the use in bread production. Cost reduction in production and inspection processes is of big concern to bread manufacturers. Inspecting bread by human eyes must be a tedious job for human inspectors, which might result in high inspection error. Hence, automated inspection systems are greatly required.

In this work, the bread image is binarized firstly. If several breads are adjacent to each other, those are separated using K-cosine corner detection algorithm. Next, defect detection is conducted by taking object's area and Hu's invariant moments.

Experimental results indicate that the proposed method can be efficiently used in the bread inspection system.

Keywords — Industrial Automation, Food inspection, K-cosine, Invariant moments

1. Introduction

Visual product quality is important to consumers. The consumer chooses a product among many competing ones, and visual quality hold often big portion in deciding which product to purchase. Because the product cannot be visually inspected before purchase, consumers will be disappointed if the product found inside the packaging does not compare well with the images displayed on the packaging.

Traditionally, visual quality product inspections have been conducted by trained human inspectors. Product inspection by human workers must be a tedious job and the workers are limited in their ability to perform objective, consistent and high speed inspection. They are prone to fatigue, and inspection performance varies over time and between individuals. It requires a cost effective and efficient automated way.

Automated visual inspection provides high-speed, cost effective, and high-quality detection of defects. Some of the most challenging industrial inspection areas include inspection of textured materials such as textile, and food inspection such as bread or cookies.

There have been many research papers on inspection of textured materials such as fabric[1-6], chemical film [7-8], wood [9-10], carpet [11], steel roll [12], and paper[13] inspection. For texture discrimination, there exists various scheme such as wavelet transform[1-3], Gabor filter[5][8], discrete cosine transform[7], and others[4][6][9].

In this paper, we focus on the food inspection, especially on breads (see Fig. 1). Being different with textured material inspection, bread inspection deals with discrete objects. In production process of breads in factory, breads may be damaged because of rolling, collision each other, and etc. It needs a different scheme with that of textured materials.



Figure 1. A Food Inspection System[14]

Shape descriptors take a role of describing a shape. The descriptors for different shapes can be different enough that the shapes can be discriminated. There are various shape descriptors and are classified by their invariance with respect to the transformations allowed in the associated shape definition [15]. Many descriptors are invariant with respect to congruency and that means translated, rotated and mirrored shape will have the same descriptor, for example, invariant moments.

Bread defect inspection consists of two-stage process: detection followed by counting. The detection process inspects the presence of defects in the bread products while the counting process counts breads in a bread image. In contrast to counting, defect detection is considered to be more meaningful, as it provides the necessary information for the grading of bread product.

This paper is organized as follows. In section 2, we label the foreground and count the number of bread in a single labeled object and separate neighboring breads into single bread. Then defect detection is conducted by invariant moments. Experimental results and the conclusion follow in sections 3 and 4, respectively.

2. Proposed Algorithm

In this section, foreground objects are classified into two types according to the number of breads in a single labeled object: single breads and multiple breads. On the conveyor, the

bread products may be connected. However, the shape of one single bread should be extracted in order to examine the defect inspection. Thus, multiple breads require separate treatment.

Once multiple breads are separated into single breads, the invariant moments of all single objects are computed for defect inspection. Among invariant moments, Hu's invariant moments [16] is used. The block diagram of the proposed algorithm is shown in Fig. 2.

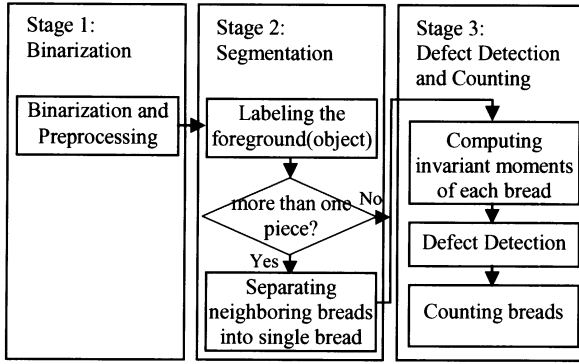


Figure 2. Block diagram of the proposed algorithm

2.1 Binarization

The binarization process distinguishes input bread image into foreground(bread) and background(non-bread) region.

$$BinaryMap(x, y) = \begin{cases} 1, (r + g + b) / 3 < \theta_1 \\ 0, \text{otherwise} \end{cases} \quad (1)$$

where r, g, b denotes the RGB values at spatial location (x, y) respectively, and the RGB values are in the range [0-255]. That is, the pixel whose intensity is less than a threshold is classified as the one in bread region. The threshold value θ_1 is dependent on experimental environments such as illumination, intensity of bread, and so on.

After thresholding, the noise which exist outside of breads on *BinaryMap* is eliminated using closing operation.

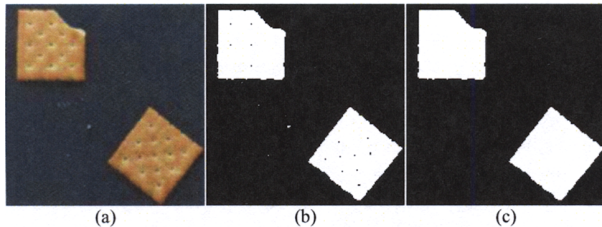


Figure 3. Binarization and preprocessing example. (a) original bread image (b) *BinaryMap*(x, y) (c) closing operation conducted image

2.2 Labeling the foreground and counting the number of bread in a single labeled object

On the *BinaryMap*, 4-connectivity Connected Component Labeling[17] is conducted to get the list of the object in the

image. This labeling provides object's attribute such as the number of pixel, size, aspect ratio, and so on.

Measuring the area of an object, the number of breads in a single connected object is determined. In this work, the number of bread on a single connected object is obtained by

$$nob = n, \text{ if } (1.05 \cdot (n-1) + \frac{(n-1)^2}{400}) < \frac{M}{A} < 1.05 \cdot n - \frac{n^2}{400}, (1 \leq n \leq 10) \quad (2)$$

where n denotes the number of bread on the object and A is the area of the reference object saved in database and M denotes the number of pixels of the single object.

2.3 Separating connected breads

This process is applied only to the objects of which *nob* is larger than one.

As mentioned in previous section, the bread products may be connected on the conveyor, which requires separate treatment for defect inspection. An example is shown in Fig. 4-(a).

In this process, it is important to detect adjacent points between connected breads. For this, several steps are needed as follows.

2.3.1 Contour Tracing

Firstly, contour tracing is conducted to detect adjacent points for all objects which the *nob* is larger than one.

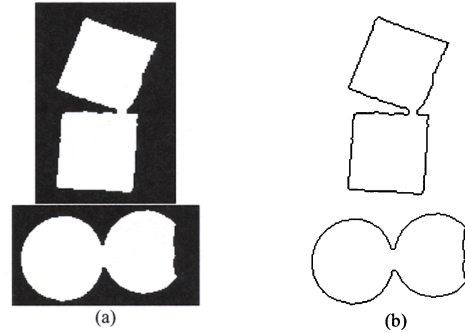


Figure 4. Contour tracing example. (a) binary images of neighboring breads (b) contour traced images of (a)

2.3.2 Detecting severe curvature point candidates

When some breads are adjacent, severe curvatures are occurred. We assume that adjacent points are among these. These severe curvatures can be detected by k -cosine.

Inesta *et al.* employ curvature to identify dominant points for polygonal approximations of real imaged objects, and defined K -curvature, K -angular bending and K -cosine [18]. The K -cosine is defined as below.

Given an object whose boundary is defined by $S = \{ P_i \mid i=1, 2, 3, \dots, m \}$, the curvature (K -cosine) of each boundary point P_i is defined as

$$C_i(K) = \cos \theta_i = \frac{\vec{a}_i(K) \cdot \vec{b}_i(K)}{\|\vec{a}_i(K)\| \cdot \|\vec{b}_i(K)\|} \quad (3)$$

As depicted in Fig.5, $\vec{a}_i(K) = \vec{P}_{i-K} - \vec{P}_i$, $\vec{b}_i(K) = \vec{P}_{i+K} - \vec{P}_i$, θ denotes the angle between $\vec{a}_i(K)$ and $\vec{b}_i(K)$, and $K \in \mathbb{N}$.

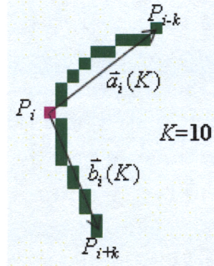


Figure 5. Curvature measurement with K -cosine

In (3), $c_i(K)$ denotes the cosine of the angle between vectors $\vec{a}_i(K)$ and $\vec{b}_i(K)$ for a given pixel P_i . K represents a region of support that is the number of pixels between the starting and ending point of a given boundary point. The K -cosine contains the curvature information such that $-1 < c_i(K) < 1$. When $\theta = 180^\circ$, $C_i(K) = -1$, indicating that the corresponding point is on a straight line. When $C_i(K)$ is close to 1, then the angle θ is approximately 0° , implying the corresponding point is on a sharp angle. Thus, when $C_i(K)$ is far from -1 , then $\vec{a}_i(K)$ and $\vec{b}_i(K)$ are in two different segments of which point P_i is a corner point.

In this work, we set k to five and the threshold for severe curvatures as -0.75 . The gaussian smoothing is conducted to $C_i(K)$ because it has much noise, before thresholding.

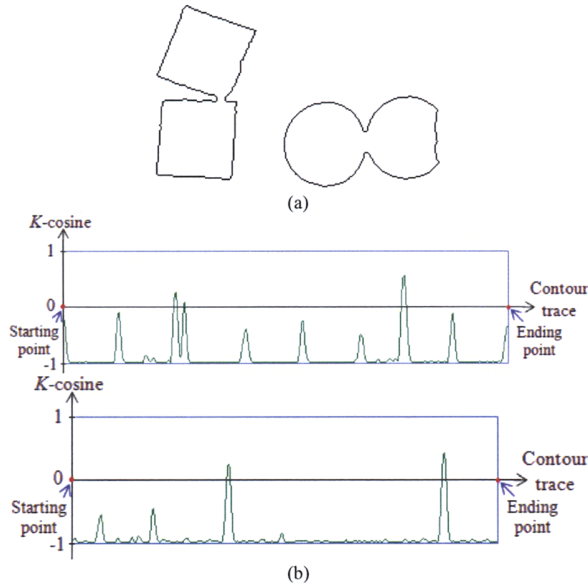


Figure 6. Detecting severe curvatures using k -cosine. (a) contour traced images (b) computed k -cosine of (a) (c) detected severe curvatures of (a) (overlaid in white rectangle).

2.3.3 Selecting points and linking two points

Once severe curvatures are detected, two of them are selected in order to separate neighboring objects into single objects. However, it is not easy to choosing two points among several points in case there are more than two severe curvatures.

In order to select two points, we propose several steps as you can see in Fig. 7. First, perform morphology erosion process till neighboring breads are separated (Fig. 7 (a)-(c)). Second, compute the center point of each separated objects. Next, connect two center points and compute middle point of the connected line. Fourth, select two points which are most close to middle point. Finally, selected two severe curvature points are linked as you can see in Fig. 7-(f).

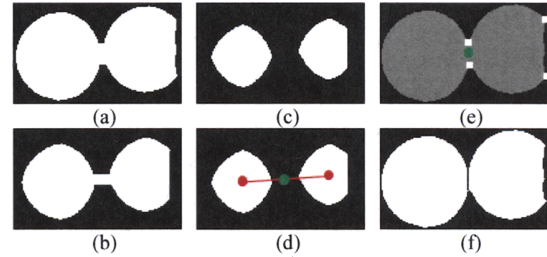


Figure 7. Edge linking example of the lower one of Fig.4-(a) (a)(b)(c) Morphological (erosion) images(conducted 1,5,9 times respectively) (d) Center points of each separated objects and middle point of connected line (overlaid in red and green circle respectively) (e) middle point of connected line is marked on severe curvatures image (f) linked result

2.3.4 Defect Detection

A uniqueness theorem [19] states that if $f(x,y)$ is piecewise continuous and has nonzero values only in a finite part of the x - y plane, moments of all orders exist, and the moment sequence m_{pq} is uniquely determined by $f(x,y)$. Conversely, m_{pq} uniquely determines $f(x,y)$. In practice, the image is summarized with functions of a few lower order moments[20].

The moment of digital image is defined by

$$m_{pq} = \sum_{y=0}^{N-1} \sum_{x=0}^{M-1} x^p y^q f(x,y) \quad (4)$$

where $0 \leq p, q$, and M, N means image width and height respectively. For example, m_{00} is the area (for binary images) or sum of grey level (for greytone images).

The central moment is defined by

$$\mu_{pq} = \sum_{y=0}^{N-1} \sum_{x=0}^{M-1} (x - \bar{x})^p (y - \bar{y})^q f(x, y) \quad (5)$$

where $\bar{x}$, $\bar{y}$ is the average x and y position, respectively:

$$\bar{x} = \frac{1}{M} \sum_{x=0}^{M-1} xf(x, y) = \frac{m_{10}}{m_{00}}, \bar{y} = \frac{1}{N} \sum_{y=0}^{N-1} yf(x, y) = \frac{m_{01}}{m_{00}} \quad (6)$$

The central moments of order up to 3 can be represented by

$$\begin{aligned} \mu_{00} &= m_{00} \\ \mu_{10} &= 0 \\ \mu_{01} &= 0 \\ \mu_{11} &= m_{11} - \bar{x}m_{01} = m_{11} - \bar{y}m_{10} \\ \mu_{20} &= m_{20} - \bar{x}m_{10} \\ \mu_{02} &= m_{02} - \bar{y}m_{01} \\ \mu_{30} &= m_{30} - 3\bar{x}m_{20} + 2\bar{x}^2m_{10} \\ \mu_{03} &= m_{03} - 3\bar{y}m_{02} + 2\bar{y}^2m_{01} \\ \mu_{21} &= m_{21} - 2\bar{x}m_{11} - \bar{y}m_{20} + 2\bar{x}^2m_{01} \\ \mu_{12} &= m_{12} - 2\bar{y}m_{11} - \bar{x}m_{02} + 2\bar{y}^2m_{10} \end{aligned} \quad (7)$$

This central moments are translational invariant.

Moments η_{pq} where $p+q \geq 2$ can be constructed to be invariant to both translation and changes in scale by dividing the corresponding central moment by the properly scaled (00)th moment, using the following formula.

$$\eta_{pq} = \frac{\mu_{pq}}{\mu_{00}^{(1+\frac{p+q}{2})}} \quad (8)$$

Hu [16] propose a set of invariant moments combining central moments, which are invariant under translation, changes in scale, and also rotation. His seven invariant moments are defined by:

$$\begin{aligned} \phi_1 &= \eta_{20} + \eta_{02} \\ \phi_2 &= (\eta_{20} + \eta_{02})^2 + 4\eta_{11}^2 \\ \phi_3 &= (\eta_{30} - 3\eta_{12})^2 + (3\eta_{21} - \eta_{03})^2 \\ \phi_4 &= (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2 \\ \phi_5 &= (\eta_{30} - 3\eta_{12})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] \\ &\quad + (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] \\ \phi_6 &= (\eta_{20} - \eta_{02})[(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] \\ &\quad + 4\eta_{11}(\eta_{30} + \eta_{12})(\eta_{21} + \eta_{03}) \\ \phi_7 &= (3\eta_{21} - \eta_{03})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] \\ &\quad + (3\eta_{12} - \eta_{30})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] \end{aligned} \quad (9)$$

Defect detection process is conducted by using Euclidean Distance[21] which is defined as follows.

$$U_d = \frac{\sum_{i=1}^4 (1 - R_i)^2}{4}, R_i = \frac{\phi_i^{Exp}}{\phi_i^{Db}} \quad (10)$$

where ϕ_i^{Db} and ϕ_i^{Exp} are the i -th Hu moments of reference and experimental object respectively and U_d is Euclidean distance between ϕ_i^{Db} and ϕ_i^{Exp} .

Only first four Hu moments are used to compute U_d in this system because ϕ_7 is skew invariant, which enables it to distinguish mirror images of otherwise identical images so it is meaningless in this kind of system. ϕ_5 and ϕ_6 also provide little information.

The objects are labeled as 'NON_DEFECT' or 'DEFECT_CANDIDATE', using the rules in (11).

$$\begin{cases} object_type = NON_DEFECT & \text{if } \begin{cases} U_d > \theta_2 \text{ or} \\ A \cdot 0.7 < M < A \cdot 1.2 \end{cases} \\ object_type = DEFECT_CANDIDATE, & \text{otherwise} \end{cases} \quad (11)$$

where A is the area of the reference object saved in database and M denotes the number of pixels of the single bread in experimental image. That is, an object is classified as defective one if the U_d is larger than threshold θ_2 or the area is out of some range. The area constraint is needed because the Hu's invariant moments are invariant to scale and the shape of objects located in brink is distorted because of radial distortion of camera.

3. Experimental Results

The proposed system was implemented using Visual Studio 6.0 (C++) under the Win32 environment. We used three types of cookie images which are bitmap format file with an image size of 1000 × 600. The experiments were conducted on a low-end PC (Intel(R) Core2 Duo CPU @ 2.66GHz).

The experimental results of defect inspection and counting breads are shown in Table 1. As shown in the table, the number of defected and non-defected breads is almost correctly counted.

Table 1. The result of defect detection and inspection

Test images	The # of breads manually counted as			The # of breads automatically counted as		
	defect	Non-defect	Total	defect	Non-defect	Total
Fig. 8-(b)	16	51	67	16	51	67
Fig. 8-(c)	13	14	27	12	15	27
Fig. 8-(d)	12	14	26	11	15	26
Fig. 8-(e)	7	16	23	7	16	23
Fig. 8-(f)	7	16	23	7	16	23

Recall and Precision are used defined as follows for measuring the accuracy of inspection.

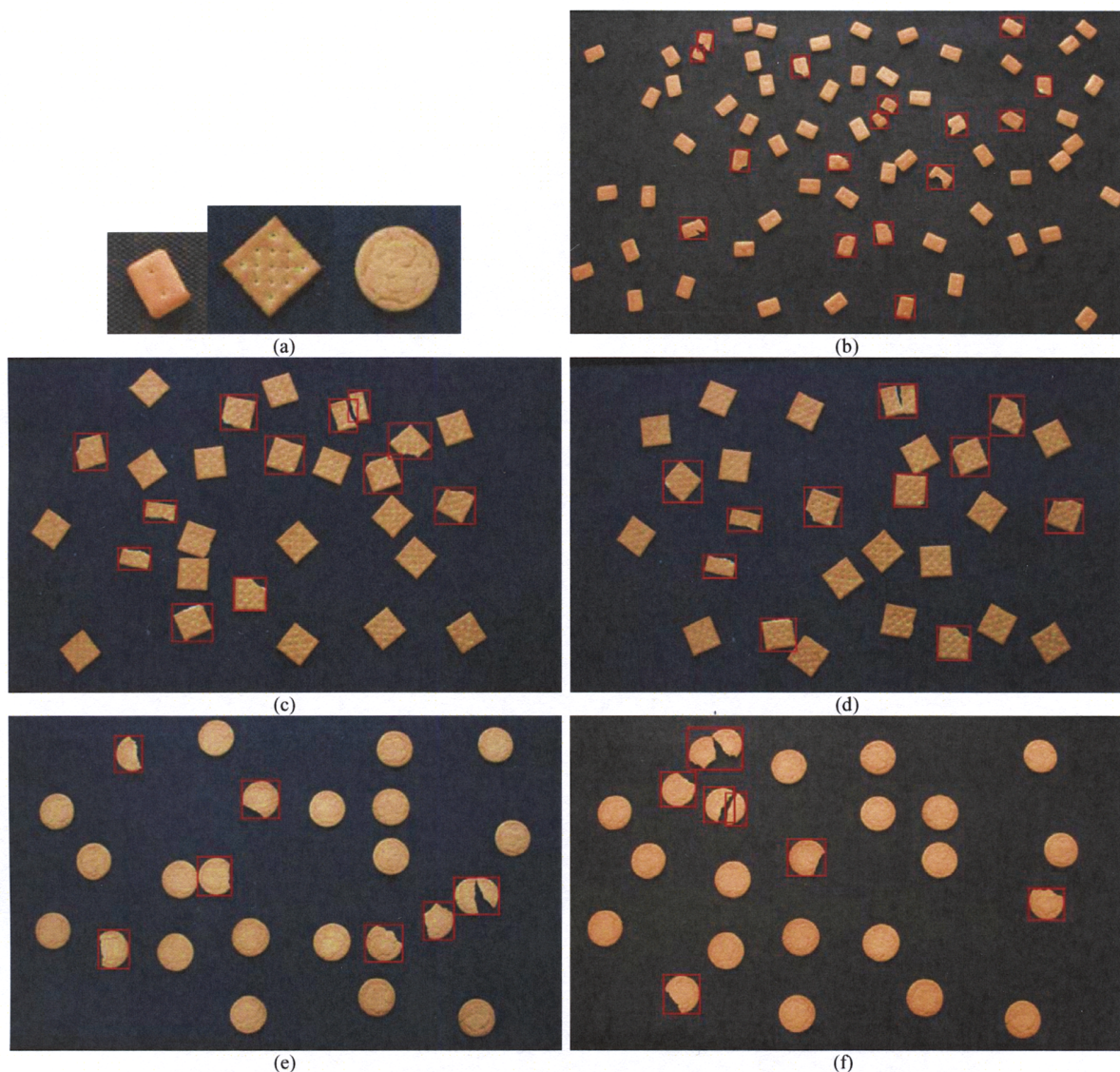


Figure 8. Bread defect detection example (a) Three kinds of reference patches saved in database (b) (c)(d) (e)(f) detected damaged breads (the Minimum Bounding Rectangles (MBRs) overlaid in red) using left, middle, and right patch of (a) respectively

$$\text{Recall} = \frac{\# \text{ of breads manually and automatically counted as defected}}{\# \text{ of breads manually counted as defected}} \quad (12)$$

$$\text{Precision} = \frac{\# \text{ of breads manually and automatically counted as defected}}{\# \text{ of breads automatically counted as defected}} \quad (13)$$

where # denotes the number.

Recall and Precision should be observed simultaneously because Recall become larger regardless of accuracy when the number of breads automatically counted as defected is larger than manually counted as defected. Precision become larger when the number of breads automatically counted as defected is smaller than manually counted as defected. Table 2 shows the Recall and Precision of fine test images.

Table 2. Recall and Precision

Test images	Recall	Precision
Fig. 8-(b)	100	100
Fig. 8-(c)	92.3	100
Fig. 8-(d)	91.7	100
Fig. 8-(e)	100	100
Fig. 8-(f)	100	100

Table 3 shows elapsed time including displaying the result image for bread inspection. The experiment was repeated three times to get averaged values for the time estimates. Average processing time is around 0.15seconds.

Table 3. Processing time for bread inspection	
Test images	Processing time (sec)
Fig. 8-(b)	0.173
Fig. 8-(c)	0.159
Fig. 8-(d)	0.171
Fig. 8-(e)	0.145
Fig. 8-(f)	0.107
Average Processing time	0.151

4. Conclusion

We propose a food inspection algorithm to offer a more cost reduction, high-speed, and high-quality detection to the manufacturers of food, such as breads. The defect detection is conducted using invariant moments. The proposed inspection scheme, including separating objects, can be performed in a short time around 0.15 seconds on a Intel Core2 Duo 2.66GHz PC for 1000×600 .

We believe that this food inspection algorithm can provide better inspection environments for food manufacturers. It can particularly improve time-efficiency because it inspects multiple objects instead of one by one. The goal of our on-going work is to construct a framework to provide robust and efficient algorithms. The framework is not necessarily limited to breads images and can be applied to other types of field food images.

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